Alejandro Rodriguez

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Education

M.S. in Computational and Applied Mathematics, May 2026, GPA: 3.8
California State University, Fullerton - Fullerton, CA

B.S. in Applied Mathematics, May 2024, GPA: 3.7
California State University, San Bernardino - San Bernardino, CA

Research Interest

Phase Array Antennas (Slotted waveguide antennas)
Stochastic Processes (Hidden Markov chains)
Combinatorics (Ramsey Theory)

Research Experience

Master's Industry Project (The Aerospace Corporation)
California State University, Fullerton January 2026 - May 2026
Low-SWaP-Cost Passive Slotted Waveguide Array Antenna Design & Simulation
Advisors: Dr. Charles H. Lee & Dr. Tien M. Nguyen

- Spearheaded a 12 GHz passive slotted waveguide array antenna research simulation project for satellite/SAR applications.
- Diagnosed critical simulation bottlenecks utilizing the MATLAB Antenna Toolbox.
- Identified computing power limitation and excitation model constraints during waveguide antenna research.
- Engineered a custom workaround leveraging multi-core parallel processing and optimizing an integrated coaxial probe feed.

Graduate Research Assistant
California State University, Fullerton January 2025 - August 2025
Modeling Uni-Cellular Organism Movement With A Hidden Markov Chain
Advisor: Dr. Nicholas Brubaker

- Developed a Hidden Markov model to characterize and predict the position and rotational behavior of a unicellular organism.
- Constructed state-transition models and boundary conditions to represent stochastic movement within a bounded domain.
- Formulated the transition matrix and performed eigenvalue analysis to characterize the system's long-term behavior.

Undergraduate Research Assistant
California State University, San Bernardino June 2023 - May 2024
Bounds on Arithmetic Rainbow Ramsey Multiplicities
Advisor: Dr. Gabriel Elvin

- Investigated extremal problems in Ramsey theory and additive combinatorics, focusing on rainbow 3-term arithmetic progressions under 3-colorings.
- Developed mathematical constructions and counting arguments to establish upper and lower bounds on maximum rainbow proportions over [n] and modulo n.
- Analyzed modular coloring constructions and proved that when n is divisible by 3, the exact maximum rainbow proportion is two thirds over modulo n.

Publications

G. Elvin, A. Gonzales, A. Rodriguez, and I. Wilbur, “Bounds on Arithmetic Rainbow Ramsey Multiplicities”, *PUMP J. Undergrad. Res*., vol. 9, pp. 325–335, Sep. 2026.

Presentations

Rodriguez, A., “Low-SWaP-Cost Passive Slot Array Antenna (PSAA) Design & Simulation Results.” *International-National University Collaboration R&D (IUCR&D) Seminar*, The Aerospace Corporation, El Segundo, Los Angeles, May 2026, Research Presentation.

Brubaker, N., Gonzalez, A., & Rodriguez, A. (Presenter). “Modeling Uni-Cellular Organism Movement Using a Hidden Markov Chain.” *2025 CSUF Summer Symposium*, California State University, Fullerton, CA, August, Poster Presentation.

Elvin, G., Rodriguez, A. (Presenter), Gonzalez, A., & Wilbur, I. “Rainbow Ramsey Theory: Quantifying Regularity in Large Structures.” *Center for Computation and Applied Mathematics Research Symposium*, California State University, Fullerton, September 2024. Poster Presentation.

Elvin, G., Rodriguez, A., Gonzalez, A., & Wilbur, I. “Rainbow Ramsey Theory: Quantifying Regularity in Large Structures.” *Meeting of the Minds Student Research Symposium*, California State University, San Bernardino, April 2024. Poster Presentation.

Elvin, G., Rodriguez, A., Gonzalez, A., & Wilbur, I. “Rainbow Ramsey Theory: Quantifying Regularity in Large Structures.” *Combinatorics Seminar for Mathematics Departmental Honors*, California State University, San Bernardino, December 2023. Research Presentation.

Elvin, G., Rodriguez, A., Gonzalez, A., & Wilbur, I. “Rainbow Ramsey Theory: Quantifying Regularity in Large Structures.” *Southern California Conference for Undergraduate Research (SCCUR)*, California State University, Fullerton, November 2023. Poster Presentation.

Elvin, G., Rodriguez, A., Gonzalez, A., & Wilbur, I. “Rainbow Ramsey Theory: Quantifying Regularity in Large Structures.” *Southern California Discrete Mathematics Symposium (SoCalDM)*, Los Angeles, CA, November 2023. Poster Presentation.

Elvin, G., Rodriguez, A., Gonzalez, A., & Wilbur, I. “Rainbow Ramsey Theory: Quantifying Regularity in Large Structures.” *Undergraduate Summer Research Program Mini Conference*, California State University, San Bernardino, August 2023. Poster Presentation.

Technical Projects

Predicting Stock Buy/Sell Signals Using a Random Forest October 2025 - December 2025

- Processed 55 years and 1.4+ GB of historical equity and metadata, converting datasets to HDF5 format to reduce I/O latency.
- Engineered 30 custom predictive features capturing market momentum, volatility, and price behavior.
- Developed a Random Forest binary classifier to predict the direction of five-day forward equity returns.
- Mitigated data leakage and look-ahead bias using three time-series cross-validation methodologies.
- Achieved 53.7% directional accuracy under rigorous backtesting constraints.

Satellite Constellation Solar Power Modeling Under Earth Occultation August 2025 - August 2025

- Investigated the effects of Earth occultation on satellite solar power generation.
- Utilized a custom multi-body orbital mechanics simulator with tensor-based constellation modeling in MATLAB.
- Modeled solar irradiance, photovoltaic power generation, and Earth’s umbra and penumbra across satellite trajectories.
- Developed a vector-geometry algorithm to dynamically detect satellite entry into and exit from Earth’s shadow.
- Simulated a 3-plane, 9-satellite low-Earth-orbit constellation over 10 days to characterize long-term power-generation behavior.

- Determined that satellites avoiding Earth occultation received 13% greater average power, while 90° panel orientations increased power generation by 14%.

Teaching & Mentoring Experience

Math Tutor - Tutoring Center November 2022 - May 2024
California State University, San Bernardino,
Courses: MATH 1301. Modeling with Functions, MATH 1402. Preparation for Calculus A, MATH 1403. Preparation for Calculus B, MATH 2210. Calculus I, MATH 2220. Calculus II, MATH 2310. Applied Linear Algebra.

Undergraduate Representative - ALRISE Alliance September 2023 - May 2024
California State University, San Bernardino

Grading Assistant - Calculus II February 2023 - May 2023
California State University, San Bernardino

Professional Development

Graduate Student Teaching and Learning Certificate Program September 2025 - November 2025
California State University, Fullerton

UCLA-CSU Summer Bridge Program July2024 - August 2024
University of California, Los Angeles

OSHA 30 Construction Safety and Health February 2017
Victor Valley Community College

Honors & Awards

Departmental Honors in Mathematics
California State University, San Bernardino | May 2024

Dean's List
California State University, San Bernardino | Spring 2022, Fall 2022, Fall 2023

Conferences Attended

Joint Mathematics Meetings (JMM) January 2024
San Francisco, CA
Travel funded by the Louis Stokes Alliances for Minority Participation (LSAMP).

California State University Mathematical Sciences Conference November 2023
Bakersfield, CA
Travel funded through faculty professional development funds.

Mathematical Association of America (MAA), Southern California–Nevada Fall Section Meeting October 2023
Channel Islands, CA
Travel funded by the Louis Stokes Alliances for Minority Participation (LSAMP).

Mathematical Association of America (MAA), Southern California–Nevada Spring Section Meeting April 2022
Pomona, CA

Skills

Programming: Python, MATLAB, pandas, scikit-learn, Matplotlib, NumPy, Seaborn, feature engineering, statistical modeling. LaTeX
Modeling: Mathematical modeling, computational simulation, numerical methods, optimization, stochastic processes, Markov chains
Engineering: MATLAB Antenna Toolbox, antenna simulation, beam pattern analysis, S-parameter analysis, orbital mechanics